Why don't Graph Transformers attend to the same token?

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Abstract

Large Language Models empirically attend heavily to the semantically meaningless <BOS> token that precedes every text sequence, creating an 'attention sink' that has revealed critical for over-mixing regularisation. Nonetheless, this phenomenon remains unexplored for Graph Transformers and cannot be spontaneously derived as permutation equivariance and full global attention suppress the asymmetric sequential pressure that drives accumulation on the first tokens. Our study thus extends a multi-layer sensitivity bound to Graph Transformers, yielding two novel results: full attention weakens sink pressure with increasing context sizes (a behaviour opposite to LLMs), and edge-conditioned gating lowers mixing pressure by reducing the spectral gap of the effective mixing matrix. We corroborate our theory with empirical observations on sink scores, representational compression, and over-smoothing. Sink rates empirically always remain below established thresholds and further decrease as node count increases: the GRIT architecture on Peptides-func notably shows 15.8% sink rate for the smallest graphs and 5.5% for the largest ones. In contrast, GraphGPS' hybrid architecture with message-passing exceeds the 30% sink rate threshold, a figure confirming the architecture-dependence of sink formation in Graph Transformers.

1 Introduction

In recent years, Transformer-based models have enabled new state-of-the-art performance in a wide variety of machine learning tasks. This advent is notably made possible by the attention mechanism which updates tokens by mixing in information from all other tokens. In natural language processing, this framework enabled the rise of auto-regressive Large Language Models (LLMs), generating human-like speech by mixing tokens iteratively to put words in relations with each other and acquire a global overview of the input.

Today, many aspects of Transformer-based models nonetheless remain poorly understood, yielding a theoretical gap that reveals both unproductive with regards to improvements of these models and dangerous given misunderstood unaligned behaviours are discovered frequently on such models. Large research efforts are thus devoted to fulfil that gap. Notably, an interesting observation is that LLMs exhibit 'attention sinks', tokens which appear to capture most of the attention. In a typical prompt on LLaMa 3.1 405B, nearly 80% of attention heads form strong sinks on the semantically meaningless <BOS> (beginning of sequence) token, an allocation which at first glance appears to waste compute. Attention sinks have been attributed to a variety of training components and many works have attempted to mitigate them. However, studying this phenomenon from another perspective yields a different paradigm: attention sinks are in reality useful from a 'mixing' perspective.

As studied in Barbero et al. (2025), the presence of attention sinks improves robustness to perturbation by slowing down the mixing of information, preventing 'over-mixing', a phenomenon covered in Dong et al. (2021), Barbero et al. (2024), and Di Giovanni et al. (2022) that triggers rank collapse, representational collapse, and over-squashing. In particular, with deeper architectures, inactivity must increase to ensure representations stay separated throughout the entire architecture. Longer contexts in LLMs also theoretically and empirically require stronger sinks.

Transformer models are not only used in LLMs. Graph Transformers also enable classifications as well as regressions on graphs, nodes, and links; they notably show very good performance on long-range graph interaction Dwivedi et al. (2022b). Contrary to LLMs, Graph Transformers do not need causal masking thanks to a lack of ordering. Nonetheless, depth still generates 'over-smoothing' in Graph Neural Networks (GNNs), where repeated aggregation causes representational collapse Zhang et al. (2023). Global attention exacerbates this risk by allowing every node to attend to every other: Veličković et al. (2024) has notably proven that for sufficiently long contexts, global attention matrices always converge to 'pure mixing'. Concentrating weight on a single node could thus regularise against representational collapse.

1.1 Research Question

While attention sinks have been thoroughly researched for LLMs and causal masking, none of the existing literature studies this phenomenon in Graph Transformers. This appears as a genuine opportunity as attention sinks cannot trivially transfer from LLMs to graphs: global attention and permutation-equivariance must bring in a different perspective. With no 'first token', it isn't clear if and where attention sinks arise in graph models. Graph topology also adds another degree of freedom absent from the simpler sequence length approach of LLMS.

This project thus investigates when and where do attention sinks form in Graph Transformers and how are they useful to maximising performance? Here, several aspects appear crucial to consider: whether sink nodes are determined by structure (e.g. by degree or spectral position), whether context length (i.e. node count) has an impact on the presence of attention sinks, and whether a virtual node connected to all other nodes can act as a designated sink analogue to <BOS> in LLMs or analogue to registers for internal computations in vision transformers (Darcet et al. (2023)). This research aims to unify our understanding of over-mixing defence across domains through a geometric lens.

2 Outline of study

Our study thus aims to extend the framework of Barbero et al. (2025) to Graph Transformers using the following approach and goals:

1. **Extend the sensitivity bound of Barbero et al. (2025) to Graph Transformers.** We derive an upper bound on end-to-end sensitivity $|\partial \mathbf{h}_j^{(L)} / \partial \mathbf{h}_i^{(0)}|$ via the Jacobian, revealing two differences with LLMs: absence of a causal mask removes asymmetric mixing pressure on nodes, and structural edge encodings reduce the spectral gap of the effective mixing matrix (per-layer inter-node influence), reducing the rate at which stacking attention layers drive representations to the mean.
2. **Empirically study the emergence of attention sinks in pure Graph Transformers at varying context lengths.** We test our result by measuring sink formation for varying graph size (which acts a context length here) with long-range interactions, a setting that strongly favours attention sinks in LLMs.
3. **Investigate how model architecture and graph structure can impact sink formation.** We compare pure Graph Transformers against hybrid architectures that combine message passing with global attention, investigate the consequence of adding a fully connected virtual node, and attempt to characterise sink identity in models containing attention sinks using structural properties like node degree and centrality.

3 Methodology

Assumptions first need to be made to best extend the results of Barbero et al. (2025) to graphs. Firstly, in terms of task at hand, graph classification appears as the closest analogue to LLMs. Both operate at the whole-input level: similarly to how LLMs uses an entire text sequence to predict its next word, a graph classification uses a whole graph to predict a given property. Secondly, given many available graph transformer architectures, we also aim to pick the closest analogue to pure LLM architectures. For this purpose, we select architectures that omit message passing whatsoever, relying solely on global self-attention and graph structure through relative positional encodings. This design choice is made as message passing introduces a hard locality bias which is the opposite of what we enforce in LLMs. It also favours over-smoothing and over-squashing which huge disadvantages for long-range interaction and which are handled trivially via direct attention.

3.1 Graph Transformer architecture and datasets

GRIT (Ma et al. (2023)) is the best architecture to extend previous attention sinks findings on LLMs to graph. Since it operates with no message passing to only rely on global self-attention, it bypasses the locality of most GNNs and popular hybrid Graph Transformer architectures such as GraphGPS (Rampášek et al. (2022)) which include local message passing and are therefore only partially comparable to LLMs. On top of that, graph structure is still injected through learned Relative Random Walk Probabilities (RRWP) that encodes pairwise relative position via random walk transition probabilities. GRIT is also state-of-the-art for many graph benchmarks, further motivating its use in our experiments. For completeness, we also compare out results against GraphGPS with Random Walk Structural Encoding form Dwivedi et al. (2021), a setup consistently at the top of Graph benchmarks. This allows us to verify our theoretical findings beyond pure Graph Transformers, enhancing the robustness of our experiments.

To evaluate sink formation, we design our experiments around datasets from Dwivedi et al. (2022a) focusing on long-range interactions where over-smoothing is dangerous and hard to avoid. This specificity thus makes it an ideal setting to test information flow and sink formation in Graph Transformers. The first dataset that we use is Peptides-func, which attempts to predict the functional classes of peptide molecules represented as graphs. Its high size variability makes it a particularly attractive candidate: with graph sizes ranging from about 30 to 300 nodes, it enables us to test sink formation with variable context sizes on the same task. Testing on different context sizes allows us to empirically study how attention sinks form across different settings and identify trends. Task freezing is also crucial: whereas language naturally converts concepts in a consistent structure, graph node interactions can vary freely across tasks. For completeness, we also evaluate our model on ZINC-12k to verify that our findings are not task-specific. ZINC-12k is also an attractive candidate as, given our theoretical results, graph regression with less context should create higher mixing pressure.

3.2 Theoretical framework

Our theoretical study adapts the multi-layer sensitivity bound of Barbero et al. (2025) to Graph Transformers. In the original LLM setting with causal masking, Barbero et al. show that with $\bar{\alpha}_{ij}^{(\ell)} = \sum_h \alpha_{ij}^{(\ell,h)} + \delta_i^j/H$, the end-to-end sensitivity is bounded by $C_{\max}^L$ times a product of lower-triangular mixing matrices. We derive the analogous bound for bidirectional, edge-conditioned graph attention (see Appendix A for the full derivation):

Theorem 3.1 *Let $C_{\max} > 0$ be the greatest Lipschitz constant of any layer, n the number of nodes, H the number of heads, and $\delta_i^j = 1$ iff $i = j$. Set $\bar{\alpha}_{ij}^{(\ell)} = \sum_h \alpha_{ij}^{(\ell,h)} + \delta_i^j$. Then:*

$$\left\| \frac{\partial \mathbf{h}_j^{(L)}}{\partial \mathbf{h}_i^{(0)}} \right\| \leq C_{\max}^L \sum_{k_1,\dots,k_{L-1} \in V} \bar{\alpha}_{j,k_{L-1}}^{(L)} \bar{\alpha}_{k_{L-1},k_{L-2}}^{(L-1)} \cdots \bar{\alpha}_{k_1,i}^{(1)} = C_{\max}^L \left[\bar{\mathbf{A}}^{(L)} \cdots \bar{\mathbf{A}}^{(1)} \right]_{ji}$$

where $\bar{\mathbf{A}}^{(\ell)} \in \mathbb{R}^{n \times n}$ is the effective mixing matrix with entries $\bar{\alpha}_{ij}^{(\ell)}$, and the sum ranges over all nodes in V. This bound differs from the LLM version as mixing matrices $B^{(\ell)}$ are full $n \times n$ and the residual term is δ_{ij} rather than δ_{ij}/H as GRIT residual connection add the full identity to the attention output before the degree scaler.

Under uniform attention ($\alpha_{ji}^{(\ell,h)} = 1/n$), the off-diagonal entries of $\bar{\mathbf{A}}^L$ scale as $[(c_A H + 1)^L - 1]/n$ where c_A absorbs the weight matrix norms (Corollary A.5 in Appendix A). This $1/n$ factor suggests that increasing the number of nodes reduces per-pair mixing pressure, opposite to LLMs where the causal mask forces accumulation on <BOS> to increase with context length.

3.3 Experimental setup

Experiments attempt to reproduce the empirical setup of Barbero et al. (2025), Ma et al. (2023), and Rampášek et al. (2022) with high fidelity. Our models thus replicate state-of-the-art architectural choices for each dataset, with the Peptides-func dataset divided into quartiles for the context-length study. Details are shown in the following table:

Model	Dataset	Dim	Heads	Layers	MPNN	PE	LR	Epochs	Batch
GRIT	Peptides	96	4	4	–	RRWP-17	3e-4	150	8
GRIT	ZINC	64	8	10	–	RRWP-21	1e-3	1500	32
GRIT w/ VNode	ZINC	64	8	10	–	RRWP-21	1e-3	1500	32
GraphGPS	ZINC	64	4	10	GINE	RWSE-28	1e-3	1500	32

Table 1: Experimental hyperparameters

Models are trained using an NVIDIA A10G GPU. In line with previous experiments in the literature with these models, training leverages the AdamW optimiser, linear warmup, cosine annealing, and gradient clipping. Together, these experiments measure the following on the test set of each dataset:

- **Sink score:** Average attention received by a node averaged over all heads (Barbero et al. (2025), Gu et al. (2024)). This enables us to define a sink rate as the fraction of (layer, head) pairs where at least one node has a sink score > 0.3, helping us detect sinks that form throughout the model. Sink scores are also correlated with degrees and betweenness centrality to reveal any potential characterisation of sink formation.
- **Representational Compression:** Following the methodology of Queipo-de Llano et al. (2025), we study norm ratio, matrix entropy, and anisotropy to track token embeddings.
- **Over-smoothing:** Dirichlet energy per layer. As shown in Cai and Wang (2020), high energy is a strong indicator of effective over-smoothing mitigation.

4 Results

Before analysing attention patterns, we verify that our models achieve competitive task performance. On ZINC-12k, GRIT reaches a test MAE of 0.065, in line with the published GRIT results of Ma et al. (2023). On Peptides-func, the per-quartile test AP ranges from 0.50 (Q4, largest graphs) to 0.64 (Q1, smallest graphs), consistent with reported baselines from Dwivedi et al. (2022b). The performance drop on larger graphs is expected given their higher classification difficulty, and confirms that the diagnostic metrics below are measured on competently trained models.

4.1 Emergence of attention sinks at varying context lengths

Recall that in Barbero et al. (2025), LLM sinks strengthen as a function of context size: models trained on short contexts (128 tokens) show near-zero sink rates, while longer contexts (1024+ tokens) produce substantially higher rates. Given a threshold of 30% for a token to be considered as a sink, this allows us to conclude that the vast majority of LLM tasks will trigger attention sinks. In line with our theorem corollary of Section 3.2 which suggests the opposite behaviour for graphs, we test sink emergence at different context length for the Peptides-func dataset on our chosen metrics from Section 3.3.

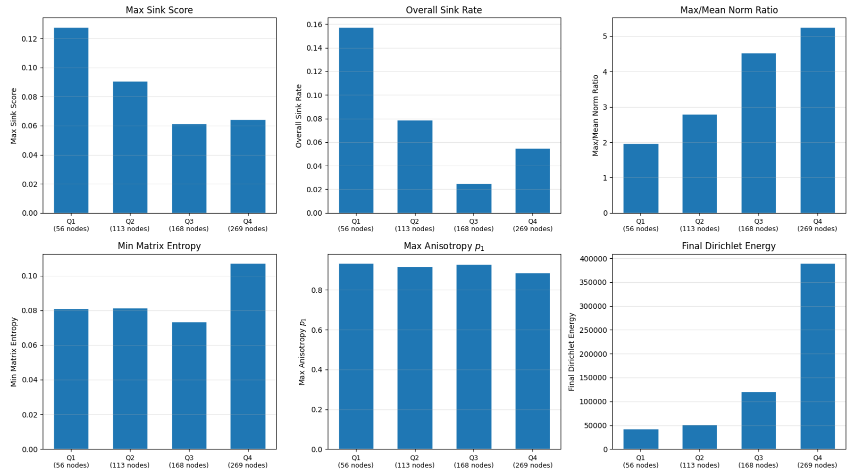

Figure 1: GRIT sink metrics for varying context length. Node count indications are averages within each quartile.

Figure 1 confirms our theoretical intuition. Max sink score drops from 0.128 to 0.064, and overall sink rate falls from 15.8% to 5.5%, below the 30% sink threshold of Barbero et al. (2025). Matrix entropy and anisotropy remain roughly stable across quartiles, suggesting that the degree of representational compression in GRIT is largely determined by architecture rather than graph size. Finally, Dirichlet energy increases with context length, indicating that larger graphs retain more feature diversity across neighbouring nodes and thus resist over-smoothing more effectively. This is consistent with the denser local structure of larger peptide graphs providing richer gradient signals through the RRWP encoding.

Norm ratio nonetheless gives a contrasting perspective: rising from 1.97 to 5.26 across quartiles, it indicates that larger graphs contain a few nodes that carry hidden-state norms far above the average. This echoes Queipo-de Llano et al. (2025) where sink tokens develop anomalously large norms. However, in GRIT, these high-norm nodes do not create attention sinks. Despite being hard to interpret with certainty, this could suggest that GRIT's specific architecture successfully decouples norm concentration from attention concentration, a theory supported by Section 4.4 which shows that the same behaviour on a different Graph Transformer architecture can be linked to attention sink formation.

We additionally replicate the perturbation sensitivity analysis of Barbero et al. (2025) by perturbing a single node's input features and measuring how the change propagates across layers as a function of graph distance. In GRIT, perturbation effects remain almost entirely contained within graph distance 1, without any visible spread. This opposes LLM results where perturbations spread far away, and is consistent with our theoretical findings, as GRIT's edge-conditioned attention concentrates mixing along graph neighbours, leaving distant nodes nearly unaffected.

This lack of attention sinks can also be verified on ZINC-12k, a dataset of smaller graphs that provides a valuable comparison on another task (molecular solubility) and should in theory maximise sink formation pressure. In this setting, sink rates indeed stay below 10%, far from the 30% threshold.

4.2 Effect of virtual nodes on attention sink formation

Another hypothesis is that adding a semantically meaningless virtual node connected to all others could become a designated sink analogue to <BOS>. As discussed in Darcet et al. (2023), adding such tokens has notably proved useful in vision transformers which also use full global attention in order to concentrate internal computations. We thus consider the setting with maximum sinking pressure and evaluate the impact of adding a virtual node on our metrics.

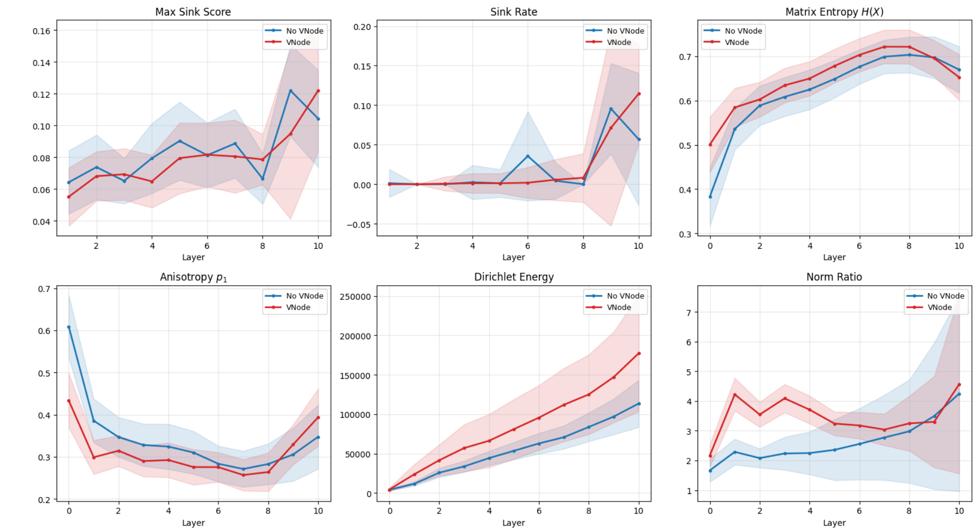

Figure 2: Layer-wise sink metrics on ZINC for GRIT with and without a virtual node

As evidenced by the figure, virtual nodes do not have a significant impact on results, consistently staying within the standard deviation of the other architecture. Furthermore, performing more analyses on the GRIT model trained with a virtual node, we discover that the virtual node is only the attention sink in 1% of cases. This suggests that, unlike for LLMs where the first token's connectedness is crucial for training, GRIT does not need fully connected nodes to prevent over-mixing. This also gives a hint at sink node characterisation, suggesting that node degree is likely not a viable indicator to predict where attention sinks will arise.

4.3 Sink node characterisation

We correlate structural properties of nodes with sink score in an attempt to characterise which nodes become sink in GRIT. Despite low sink rates, understanding where sink form can help pinpoint key properties that drives sink formations and backs our theoretical findings.

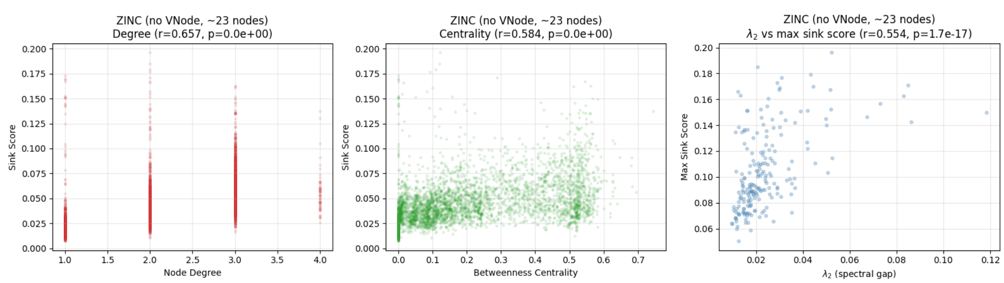

Figure 3: Correlation between sink scores and different structural properties

Even in the setting where sink rate was the highest for GRIT, results seem to point towards behaviours that generalise weakly. Regarding node degree, higher degree nodes show higher sink scores. This is intuitive: since RRWP encodes random walk probabilities which visit high-degree nodes more often, their positional encoding makes them better candidates for sink formation. At the graph-level, the same phenomenon seems to appear through a spectral gap analysis: well-connected graphs consistently show higher sink scores than poorly connected ones. Looking at betweenness centrality, we observe another weak behaviour: bridge nodes (which should concentrate information flow given long range interactions) develop higher sink rates. This is consistent with a "routing" interpretation of sinks, and is further supported by the fact that leaves (with a betweenness centrality score of 0) reveal much lower sink scores than other nodes.

4.4 Message-passing creates sink formation pressure

In contrast to task independence shown in Section 4.1, attention sink formation is highly dependent on architecture. Comparing our pure graph transformer architecture with a hybrid GNN that includes a message-passing mechanism, we confirm our theoretical findings that only pure Graph Transformers prevent sink formations and that message-passing creates a conditioning akin to LLMs pushing for the emerge of attention sink.

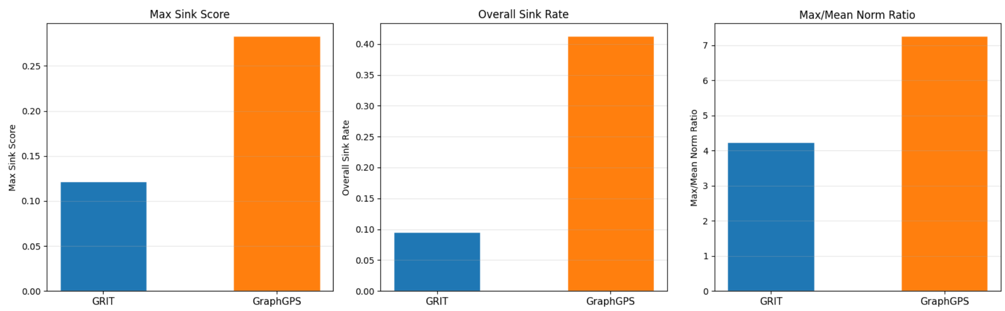

Figure 4: Evaluation of GraphGPS against GRIT on ZINC

Comparing both state-of-the-art models on the same ZINC dataset, we can isolate the effect of model architecture on sink formation. GraphGPS experiments notably yield a max sink score of 0.28 and a sink rate above 40%, above the threshold set by Barbero et al. (2025). This is very interesting as it shows the extent of our theoretical findings. Adding message-passing on the same graph and task produces qualitatively different attention patterns.

Critically, relative to GRIT, GraphGPS exhibits only 65% higher max/mean norm ratio but a $> 4\times$ higher sink rate. In line with Queipo-de Llano et al. (2025), this suggests that norm concentration observed for GRIT in Section 4.1 should have translated into higher attention sinks but that GRIT's architecture decoupled norm growth from attention concentration. GraphGPS' message-passing mechanism and softmax attention enables high-norm nodes to dominate attention, yielding attention sinks similarly to LLMs.

5 Discussion

Following the approach of Barbero et al. (2025) for LLMs, rather than limiting experiments to quantification of attention sink emergence in Graph Transformers, we attempt to describe *why* attention sinks aren't useful in our setup. Contrary to what one could infer, we thus conclude that attention sinks are not an intrinsic consequence of training attention modules. They are rather a mechanical response to how much persistent structural information the attention mechanism has access to. In LLMs, attention scores are computed purely from content representations that are themselves the product of prior mixing. As depth increases, these representations converge and attention becomes uniform, forcing the model to develop sinks as an emergent defence. In GraphGPS, the MPNN component introduces local mixing creates enough pressure for sinks to exceed the 30% threshold. In GRIT, the pairwise edge residual stream provides each layer with structural information that is never destroyed by node-level over-mixing, maintaining non-uniform attention throughout depth and lowering sink formation pressure.

This contrast suggests that attention sink formation is inversely proportional to the amount of structural information that can travel throughout the network. Model architectures that overly rely on content representation that are potentially over-mixed are sensitive to attention sinks. On the other hand, designs that maintain independent structural information channels like edge-gating are more robust to attention sinks. Overall, this reasoning allows us to infer that the emergence of attention sinks in LLMs is thus more of consequence of its causal and minimal relational design rather than a fundamental property of the Transformer architecture.

This conclusion naturally connects to several concepts of the geometric deep learning textbook (Bronstein et al. (2021)). Course contents frame Transformers as *attentional GNNs over a complete graph*. Our results support that perspective, while also revealing that edge-conditioned architectures like GRIT affect over-mixing dynamics. By injecting priors into attention modules, they manage to break the symmetry of the complete graph and alter information flow. Furthermore, permutation equivariance of graph models removes a critical precondition of attention sinks. By removing causality, there is no 'first node' onto which attention should accumulate. Finally, our spectral gap analysis in Corollary A.7 builds on the spectral graph theory covered in the course, where the Laplacian eigenvalues characterise frequencies on the graph and the Dirichlet energy measures signal smoothness.

5.1 Generalisation of our findings

Our findings give indications for attention sink behaviours in transformer architectures that generalise at different levels:

1. **Graph Transformers (Broad scope).** The multi-layer bound and dilution of sinking pressure with increasing context size from Section 3.2 should generalise across all graph transformers thanks to their full global attention. Our empirical results corroborate these indications as both GraphGPS and GRIT show sink rates that are far below LLMs.
2. **Architectures with pairwise structural bias (Narrow scope).** Focusing on pairwise information, structural bias should reduce spectral gap that in turn lowers sink rates from Corollary A.7. While specific mechanisms can differ (e.g. element-wise gating or additive bias), reduction in spectral gap should make sink formation pressure decrease mechanically, as shown in our empirical findings of Section 4.3.

5.2 Limitations of our study

Firstly, our limited architecture focus—extensively testing only on GRIT and comparing with GraphGPS—is a limitation that hinders the robustness of our empirical verification as well as the generalisation power of our theoretical findings. On the experimental side, GRIT is one specific edge-conditioned Graph Transformer, but there are many more state-of-the-art architectures (e.g EGT, Graphormer, SAN) that would also be very interesting to test (Yuan et al. (2025)). This could allow us to test our theoretical hypotheses on more examples and possibly discover behaviours that aren't observable from a single data point. Furthermore, the lack of causal ablation in our experiments also prevents us from isolating sink formation phenomena with great robustness. Doing step-by-step ablations of the GRIT architecture could allow us to test Theorem A and Corollary A.7 with greater depth. On the theoretical side, partially focusing on GRIT components limits Section 5.1's generalisations, as modules that power other popular GNNs but that aren't part of GRIT are omitted by our theoretical framework.

Secondly, our sensitivity bound adapted from Barbero et al. (2025) has limited expressivity as it is an upper bound inheriting looseness from the triangle inequality. It accurately describes qualitative behaviour of sink formation, but it does not give us quantifiable results. This is problematic in some cases as small architectural variations may have a larger impact on sink rates than what our bound loosely indicates, obscuring our interpretation. On top of that, our theoretical framework assumes that we only freeze attention weights; in GRIT, edge scores simultaneously determine both attention logits and edge-value contribution to node updates. Freezing them thus suppresses two correlated channels of information flow rather than one, loosening our bound and potentially underestimating true sensitivity.

Lastly, the narrow dataset scope (testing only on the ZINC and Peptides-func dataset) is a last computational limitation of our experiments. While both dataset have varying graph sizes, both tasks belong to the same domain (molecular chemistry), focus on long range interactions, and have close to uniform degree distributions. Indeed, it is possible that different datasets such as social networks or citation graphs with structures and feature that differ greatly with molecular graphs exhibit different sink pattern. This once again hinders generalisation and would be a crucial evaluation to ensure the robustness of our findings.

6 Outlook

Additional studies can be conducted to obtain more evidence backing our theoretical findings and better understand attention sink formation in the general context of transformers. This is notably crucial in a context where complex architectures and hyperparameters makes it hard to establish direct cause-effect relation in empirical studies.

A reasonable starting point for subsequent research is an ablation study of GRIT components, to truly break down the effect of each component of the pure Graph Transformer architecture on sink rates. Since our theory attributes sink suppression to GRIT's edge gating, one could replace edge-conditioned scores with standard dot-product attention to see how strongly Corollary A.7 empirically affects anti-sink pressure. On top of that, a study categorising specific Graph Transformer components could help us go beyond the narrow component scope of our theoretical and empirical findings, to establish behaviours that can be generalised for any architecture.

Another direction could also focus on linking sink formation and task accuracy. By artificially enforcing sink suppression or emergence on the same architecture through regularisation terms on metrics such as entropy, one could attempt to evaluate the optimality of attention sinks in GRIT. Naturally, with more computational resources, these new experiments should also extend to different graph prediction datasets, to evaluate the emergence of sink and their impact on test accuracies for tasks with shorter interactions and graph structure which are highly heterogeneous such as social networks. This would close the loop between the theoretical mixing analysis and practical model quality by allowing us to observe how the lack of sinks in GRIT impacts test accuracy.

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A Sensitivity Bound for Edge-Conditioned Graph Transformers

We adapt the sensitivity bound of Barbero et al. (2025) (Theorem 3.2) to bidirectional, edge-conditioned graph attention as implemented in GRIT (Ma et al., 2023). The original bound applies to causal Transformers with lower-triangular mixing matrices and a residual term δ_{ij}/H. In this appendix, we derive an analogous result for the GRIT model, where the mixing matrices are full $n \times n$ and the residual term is δ_{ij} rather than δ_{ij}/H.

A.1 Setup

Each node u has a representation $\mathbf{h}_u^{(\ell)} \in \mathbb{R}^d$ at layer $\ell \in \{0, \dots, L\}$, and each pair (i, j) carries an edge representation $\mathbf{e}_{ij}^{(\ell)} \in \mathbb{R}^d$ initialised from RRWP and updated via residual connections. At each layer, GRIT computes edge-conditioned attention weights $\alpha_{ij}^{(\ell,h)}$, aggregates values $\mathbf{v}_j^{(h)} = [W_V^{(\ell)}\mathbf{h}_j^{(\ell)}]_h$ with an output projection $W_O^{(\ell)}$, applies learned degree scalers $D_i^{(\ell)}$, and passes through a Lipschitz FFN $\psi^{(\ell)}$, both with residual connections. We also make the following assumptions:

- **(A1) Frozen attention path.** Attention weights $\alpha_{ij}^{(\ell,h)}$ and edge scores $\hat{\mathbf{e}}_{ij}^{(\ell,h)}$ are constants.
- **(A2) Boundedness.** The degree scaler satisfies $\|D_i^{(\ell)}\| \leq C_{\text{DS}}$ and the FFN is Lipschitz-bounded.

A.2 Single-layer bound

Lemma A.1 (Single-layer Jacobian bound)

$$\left\| \frac{\partial \mathbf{h}_j^{(\ell+1)}}{\partial \mathbf{h}_i^{(\ell)}} \right\| \leq C_{\text{FFN}}^{(\ell)} \cdot B_{ji}^{(\ell)}$$

where $C_{\text{FFN}}^{(\ell)} = \text{Lip}(\psi^{(\ell)}) + 1$ *and* $B_{ji}^{(\ell)} = c_A^{(\ell)} \sum_{h=1}^{H} \alpha_{ji}^{(\ell,h)} + \delta_{ij}$ *with* $c_A^{(\ell)} = C_{\text{DS}} \cdot \|W_O^{(\ell)}\| \cdot \max_h \|W_V^{(\ell,h)}\|$.

Proof A.2 *The FFN residual gives* $\|J_\psi + I_d\| \leq C_{\text{FFN}}^{(\ell)}$. *Given frozen attention, the only pathway from* $\mathbf{h}_i^{(\ell)}$ *to the attention output is through the value projection:* $\partial \mathbf{a}_j^{(\ell,h)} / \partial \mathbf{h}_i^{(\ell)} = \alpha_{ji}^{(\ell,h)} W_V^{(\ell,h)}$.

Passing through W_O*, the degree scaler, and adding the residual* δ_{ij} *gives* $B_{ji}^{(\ell)}$ *by sub-multiplicativity. The product of both factors yields the result.*

A.3 Multi-layer bound

Theorem A.3 (Sensitivity bound for GRIT) *With* $C_{\max} = \max_\ell C_{\text{FFN}}^{(\ell)}$*:*

$$\left\| \frac{\partial \mathbf{h}_j^{(L)}}{\partial \mathbf{h}_i^{(0)}} \right\| \leq C_{\max}^L \cdot \left[B^{(L-1)} \cdots B^{(0)} \right]_{ji}$$

where each $B^{(\ell)} \in \mathbb{R}^{n \times n}$ *has entries* $B_{ji}^{(\ell)} = c_A^{(\ell)} \sum_h \alpha_{ji}^{(\ell,h)} + \delta_{ij}$.

Proof A.4 *Expanding via the chain rule over intermediate nodes* $k_0 = i$, $k_L = j$ *and applying Lemma A.1:*

$$\left\| \frac{\partial \mathbf{h}_j^{(L)}}{\partial \mathbf{h}_i^{(0)}} \right\| \leq \sum_{k_1, \dots, k_{L-1}} \prod_\ell C_{\text{FFN}}^{(\ell)} B_{k_{\ell+1}, k_\ell}^{(\ell)} = \left(\prod_\ell C_{\text{FFN}}^{(\ell)} \right) \left[B^{(L-1)} \cdots B^{(0)} \right]_{ji}$$

This differs from Barbero et al. (2025) as mixing matrices $B^{(\ell)}$ are full $n \times n$ and the residual term is δ_{ij} rather than δ_{ij}/H. We note that assumption (A1) freezes the edge scores, so the bound does not directly capture the edge residual stream's role in maintaining structured attention. Our interpretation in the main text that edge gating reduces mixing pressure is supported by the bound through the non-uniform values of $\alpha_{ji}^{(\ell,h)}$ that edge conditioning produces, rather than through an explicit edge-channel derivative.

A.4 Corollaries

Corollary A.5 (Uniform attention baseline and $1/n$ scaling) *If $\alpha_{ji}^{(\ell,h)} = 1/n$ for all i, j, h, ℓ and layer constants are identical, then $B = (c_A H/n)\, J + I$ where $J = \mathbf{1}\mathbf{1}^\top$, with eigenvalues $\mu_1 = c_A H + 1$ and $\mu_k = 1$ for $k \geq 2$. For $i \neq j$:*

$$[B^L]_{ji} = \frac{(c_A H + 1)^L - 1}{n}$$

This decreases *with n: more nodes reduce per-pair mixing pressure.*

Proof A.6 *J has eigenvalue n for $\mathbf{1}/\sqrt{n}$ and 0 otherwise, giving $\mu_1 = c_A H + 1$, $\mu_{k\geq 2} = 1$. Via the spectral decomposition: $[B^L]_{ji} = (c_A H + 1)^L/n + (\delta_{ij} - 1/n)$.*

Corollary A.7 (Edge gating reduces spectral gap) *Suppose each head concentrates a fraction $(1 - \varepsilon)$ of its weight within the RRWP horizon (k local neighbours) and spreads ε approximately uniformly over the remaining nodes. Then $B = c_A H(1 - \varepsilon)\tilde{A}_{\text{local}} + (c_A H\varepsilon/n)J + I$ where $\tilde{A}_{\text{local}}$ is the row-normalised local adjacency. The spectral gap becomes:*

$$\mu_1 - \mu_2 = c_A H \left[1 - (1 - \varepsilon)\tilde{\lambda}_2\right]$$

where $\tilde{\lambda}_2$ is the second eigenvalue of $\tilde{A}_{\text{local}}$. For well-connected local neighbourhoods where $\tilde{\lambda}_2 \approx 1$, this reduces to $\approx c_A H\varepsilon \ll c_A H$.

Proof A.8 *The leading eigenvalue remains the same: $\mu_1 = c_A H(1 - \varepsilon) + c_A H\varepsilon + 1 = c_A H + 1$. The second eigenvalue is $\mu_2 = c_A H(1 - \varepsilon)\tilde{\lambda}_2 + 1$ (the global J component contributes only to the leading eigenvector). The gap follows by subtraction.*